

Billiards as a Typed Geometry of Observation Laboratory

Collision reduction, symbolic channels, and spectral quantities

Stas

Independent Research Program; ORCID: 0009-0000-2294-705X

Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0. It is an application-layer reference migrated to GO Core v0.2. It does not claim a new theorem in billiard dynamics, quantum chaos, or inverse spectral geometry. The new results internal to this note are typing, factorization, identifiability, and benchmark statements for explicitly declared observation channels.

Abstract

Let $\Omega \subset \mathbb{R}^2$ be a bounded billiard table. We separate four objects that are often conflated: the continuous billiard flow, its collision-section reduction, a deterministic observation of collision data, and a finite or stochastic recorded channel. On the regular collision space

$$\mathcal{M}_\partial = \{(s, \varphi) : s \in \partial\Omega_{\text{reg}}, |\varphi| < \pi/2\},$$

the billiard map is invertible almost everywhere and preserves

$$d\mu_\partial = \frac{\cos \varphi}{2|\partial\Omega|} ds d\varphi = \frac{1}{2|\partial\Omega|} ds dp, \quad p = \sin \varphi.$$

Therefore the impact map itself is not an information-loss map. Loss occurs only after a declared projection, partition, finite horizon, missing-event policy, or stochastic channel. For a known strictly convex table, two consecutive regular impact positions recover the outgoing direction, whereas one impact position does not. Finite symbolic words are necessarily non-injective on a continuum of initial states, but an infinite coding may be generating; marginal symbol entropy is neither entropy rate nor Kolmogorov-Sinai entropy. For the quantum layer, Laplacian eigenvalues λ_n have dimension L^{-2} , while physical energies are

$$E_n = \frac{\hbar^2}{2m} \lambda_n.$$

Rigid-motion invariance, similarity covariance, Weyl recovery of area, and isospectral non-identifiability are stated with their hypotheses. The result is a strict observation interface and reproducible benchmark, not a new billiard theory.

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1 Scope, conventions, and claim firewall

1.1 Regular table class

Unless a stronger hypothesis is stated, $\Omega \subset \mathbb{R}^2$ is bounded, connected, and open, with boundary equal to a finite union of piecewise C^3 arcs. The set of corners is $\partial\Omega_{\text{sing}}$; the regular part is $\partial\Omega_{\text{reg}}$. A trajectory hitting a corner, grazing the boundary, or accumulating infinitely many collisions in finite time is outside the default regular orbit class. Convexity, strict convexity, dispersing curvature, focusing curvature, and finite horizon are additional hypotheses, never defaults.

The dimensional model has a declared speed $c_0 > 0$. A regular hidden state is

$$z = (q, u) \in X := \Omega \times \mathbb{S}^1, \quad v = c_0 u, \quad \|u\| = 1.$$

Between collisions,

$$\dot{q} = c_0 u, \quad \dot{u} = 0. \tag{1}$$

At $q \in \partial\Omega_{\text{reg}}$, let $n(q)$ be the inward unit normal. If $u^- \cdot n < 0$, elastic specular reflection is

$$u^+ = u^- - 2(u^- \cdot n)n. \tag{2}$$

The convention is part of the model: changing to an outward normal changes the inequalities but not the physical reflected vector.

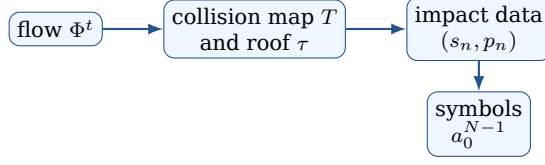
Boundary 1.1 (No universal billiard claim). Properties of circular, elliptic, dispersing, polygonal, and stadium billiards are not interchangeable. In particular, ergodicity, mixing, positive Lyapunov exponents, decay of correlations, and integrability require separate table classes and hypotheses.

1.2 Typed observation chain

For a protocol record λ , the complete chain is

$$X \xrightarrow{S} \mathcal{M}_\partial \xrightarrow{\mathcal{O}_\lambda} Z_\lambda \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_\lambda \mid Z_\lambda) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}. \tag{3}$$

Here S is collision-section reduction, not a sensor; $\mathcal{O}_\lambda$ is deterministic; $\mathcal{K}_\lambda$ is either a declared stochastic channel or the identity kernel; and $\hat{\theta}_\lambda$ is an estimator or direct readout. The continuous flow is related to the collision map by a roof function and is not identified with the discrete map.



2 Collision space, singularities, and invariant measure

Parameterize each regular boundary component by arclength s , with positively oriented unit tangent $t(s)$. For an outgoing unit vector write

$$u = \cos \varphi n(s) + \sin \varphi t(s), \quad \varphi \in (-\pi/2, \pi/2),$$

and set $p = \sin \varphi \in (-1, 1)$.

Definition 2.1 (Regular collision space and map).

$$\mathcal{M}_\partial = \{(s, \varphi) : s \in \partial\Omega_{\text{reg}}, |\varphi| < \pi/2\}.$$

Let $\mathcal{S}_1$ contain states whose next collision is a corner, grazing, non-unique, or undefined. The collision map is

$$T : \mathcal{M}_\partial \setminus \mathcal{S}_1 \longrightarrow \mathcal{M}_\partial, \quad T(s, \varphi) = (s', \varphi').$$

The roof function

$$\tau(s, \varphi) = \frac{\|q(s') - q(s)\|}{c_0} > 0$$

is the physical flight time to the next collision.

Proposition 2.2 (Regular invertibility and invariant measure). *For the reversible elastic billiard, T is invertible away from forward and backward singularity sets. It preserves the two-form*

$$\omega_\partial = \cos \varphi \, ds \wedge d\varphi = ds \wedge dp$$

and the normalized probability measure

$$d\mu_\partial = \frac{\cos \varphi}{2L_\partial} \, ds \, d\varphi = \frac{1}{2L_\partial} \, ds \, dp, \quad L_\partial := |\partial\Omega|. \quad (4)$$

Proof. Specular reflection is time reversible, so reversing the outgoing velocity at a regular collision supplies the inverse branch. The preservation of ω_∂ is the standard Poincare-section restriction of Liouville measure for billiards. Normalization is direct:

$$\frac{1}{2L_\partial} \int_{\partial\Omega} ds \int_{-\pi/2}^{\pi/2} \cos \varphi \, d\varphi = 1.$$

□

Boundary 2.3 (Reduction is not observation loss). The full pair (s_n, p_n) evolves under an almost-everywhere invertible, measure-preserving map. Calling T itself a compression is incorrect. The flow requires the roof τ , and information loss must be assigned to a subsequent observation channel.

3 Impact-position observation and identifiability

Define the instantaneous position readout

$$\pi_s : \mathcal{M}_\partial \rightarrow \partial\Omega_{\text{reg}}, \quad \pi_s(s, p) = s.$$

This map is non-injective because all $p \in (-1, 1)$ share the same output at fixed s . Sequential observations have different identifiability.

Proposition 3.1 (Two-impact reconstruction). *Let Ω be a known strictly convex table with C^1 boundary. For two distinct consecutive regular impact points*

$$q_n = q(s_n), \quad q_{n+1} = q(s_{n+1}),$$

the outgoing direction and collision coordinate at q_n are uniquely

$$u_n^+ = \frac{q_{n+1} - q_n}{\|q_{n+1} - q_n\|}, \quad p_n = u_n^+ \cdot t(s_n). \quad (5)$$

Thus the two-step position observation (s_n, s_{n+1}) reconstructs (s_n, p_n) on the stated regular class.

Proof. Free flight is a straight segment and strict convexity makes the open chord lie in Ω . Its oriented endpoints therefore determine the unit direction. Projection on the declared tangent gives p_n . $\square$

Remark 3.2. Unknown geometry, unordered impacts, missed impacts, coincident points, finite spatial resolution, non-convex visibility, and stochastic misclassification enlarge the observation fiber. The proposition does not survive those changes without additional hypotheses.

For a horizon N , useful deterministic channels are

$$\mathcal{O}_{\text{full},N}(z) = (s_k, p_k)_{k=0}^{N-1}, \quad (6)$$

$$\mathcal{O}_{\text{pos},N}(z) = (s_k)_{k=0}^{N-1}, \quad (7)$$

$$\mathcal{O}_{\alpha,N}(z) = (a_k)_{k=0}^{N-1}, \quad a_k = \alpha(s_k), \quad (8)$$

where $\alpha : \partial\Omega \rightarrow \{1, \dots, m\}$ is a declared measurable partition with a boundary-assignment rule.

4 Finite symbolic channels and entropy semantics

Proposition 4.1 (Finite-horizon non-injectivity). *Let $U \subset \mathcal{M}_\partial$ be uncountable and suppose $\mathcal{O}_{\alpha,N}$ is defined on U , with $m < \infty$ and $N < \infty$. Then*

$$\mathcal{O}_{\alpha,N} : U \rightarrow \{1, \dots, m\}^N$$

is not injective.

Proof. The codomain contains at most m^N words, while U is uncountable. $\square$

This proposition does not imply that every infinite symbolic coding is lossy. A generating partition may identify almost every orbit from its bi-infinite itinerary, modulo partition boundaries.

Let $Z_0 \sim \mu_\partial$, $Z_k = T^k Z_0$, and $A_k = \alpha(\pi_s Z_k)$. The process is stationary. For a logarithm base $b > 1$, define

$$H_b(A_0) = - \sum_{j: p_j > 0} p_j \log_b p_j, \quad (9)$$

$$H_b(A_0^{n-1}) = - \sum_{w: P_n(w) > 0} P_n(w) \log_b P_n(w), \quad (10)$$

$$h_b(T, \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} H_b(A_0^{n-1}), \quad (11)$$

where the last limit exists for a stationary finite-alphabet process. The Kolmogorov-Sinai entropy is

$$h_{\mu_\partial, b}(T) = \sup_{\alpha} h_b(T, \alpha).$$

Equality for one partition requires a generating hypothesis.

Boundary 4.2 (Marginal entropy is not dynamical entropy). The empirical quantity

$$\hat{H}_b = - \sum_j \hat{p}_j \log_b \hat{p}_j$$

uses only one-symbol frequencies. It ignores order, is partition-dependent, and is neither an entropy-rate estimate nor a KS-entropy estimate without an explicit stochastic and asymptotic protocol.

5 Exact integrable reference: the disk

Let $\Omega = B_R(0)$, $L_\partial = 2\pi R$, and retain $p = \sin \varphi$.

Theorem 5.1 (Disk collision map). *On the regular disk collision space,*

$$T_R(s, p) = (s + 2R \arccos p \pmod{2\pi R}, p). \quad (12)$$

The reduced angular-momentum coordinate

$$j_z := q \times u = Rp$$

is conserved and has the dimension of length. The physical specific angular momentum is $q \times v = c_0 j_z$. The rotation number and roof time are

$$\rho(p) = \frac{\arccos p}{\pi} \pmod{1}, \quad \tau_R(p) = \frac{2R\sqrt{1-p^2}}{c_0}. \quad (13)$$

Proof. At a collision, reflection reverses the normal component of u and preserves its tangential component p . Since $q = Re_r$, $q \times u = R(u \cdot e_\theta) = Rp$, while multiplication by c_0 restores the physical specific angular momentum. The chord subtends the central angle $\pi - 2\varphi = 2\arccos p$, giving equation (12). Its length is $2R \cos \varphi = 2R\sqrt{1-p^2}$, which gives the roof. $\square$

The disk is an exact benchmark, not evidence that a generic billiard is integrable. For an ellipse, tangency to a member of a confocal caustic family is a classical invariant on regular trajectories; this statement is retained only as a standard example and is not extended to arbitrary convex tables.

6 Chaotic tables and finite-time diagnostics

The Bunimovich stadium is a standard focusing table for which ergodicity and hyperbolic behavior are classical results under the stadium hypotheses. Those results do not justify reading a Lyapunov exponent from an arbitrary two-trajectory plot.

For z outside singularities and a tangent vector $\xi \neq 0$, the mathematical exponent is

$$\lambda_+(z, \xi) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \frac{\|DT_z^N \xi\|}{\|\xi\|}, \quad (14)$$

when the limit exists. A finite-window diagnostic must instead be declared as

$$\hat{\lambda}_{N_0, N_1} = \frac{\sum_{k=N_0}^{N_1} w_k(k - \bar{k}) \log(d_k/d_*)}{\sum_{k=N_0}^{N_1} w_k(k - \bar{k})^2}, \quad (15)$$

where d_k is computed in a declared collision-space metric, $d_* > 0$ is a reference distance, $w_k \geq 0$, and $\bar{k} = \sum w_k k / \sum w_k$. The fit must state the window, renormalization rule, saturation cutoff, singularity handling, and uncertainty model. Without those fields, $\hat{\lambda}$ is only a plot slope.

Boundary 6.1 (No chaos from one positive fit). A positive finite-window slope does not prove a positive asymptotic Lyapunov exponent, ergodicity, mixing, Bernoulli behavior, or positive KS entropy. These are distinct claims.

7 Quantum billiards: operator, dimensions, and spectrum

For a bounded open Ω , define the Dirichlet quadratic form

$$q_\Omega[\psi] = \int_\Omega \|\nabla \psi\|^2 dx, \quad \psi \in H_0^1(\Omega).$$

Its Friedrichs extension is the nonnegative self-adjoint Dirichlet Laplacian $-\Delta_\Omega^D$. It has compact resolvent and spectrum

$$0 < \lambda_1(\Omega) \leq \lambda_2(\Omega) \leq \dots, \quad \lambda_n \rightarrow \infty, \quad [\lambda_n] = L^{-2}, \quad (16)$$

with multiplicity.

Proposition 7.1 (Laplacian-to-energy bridge). *For a nonrelativistic particle of declared mass $m > 0$,*

$$\hat{H}_\Omega = -\frac{\hbar^2}{2m} \Delta_\Omega^D, \quad E_n = \frac{\hbar^2}{2m} \lambda_n.$$

Thus λ_n is not an energy. The bridge has SI dimension

$$\frac{[\hbar]^2 [\lambda_n]}{[m]} = \frac{(ML^2T^{-1})^2 L^{-2}}{M} = ML^2T^{-2}.$$

Proposition 7.2 (Rigid invariance and scale covariance). *For a Euclidean isometry g and scale $a > 0$,*

$$\lambda_n(g\Omega) = \lambda_n(\Omega), \quad \lambda_n(a\Omega) = a^{-2} \lambda_n(\Omega). \quad (17)$$

Consequently, in two dimensions,

$$\tilde{\lambda}_n(\Omega) := |\Omega| \lambda_n(\Omega) \quad (18)$$

is invariant under similarities.

Proof. Rigid motions induce a unitary pullback on L^2 . Under $x = ay$, the Laplacian becomes $a^{-2}\Delta_y$, while $|a\Omega| = a^2|\Omega|$. $\square$

Theorem 7.3 (Weyl area readout). *Let $N_\Omega(\Lambda) = \#\{n : \lambda_n < \Lambda\}$. For a planar open set of finite area,*

$$N_\Omega(\Lambda) \sim \frac{|\Omega|}{4\pi} \Lambda \quad (\Lambda \rightarrow \infty). \quad (19)$$

Hence the full Dirichlet spectrum determines area. Boundary-length terms require additional boundary regularity and a stated asymptotic theorem; they are not inferred from equation (19) alone.

Theorem 7.4 (Spectral non-identifiability). *There exist noncongruent planar domains with the same Dirichlet spectrum, including multiplicities. Therefore*

$$\mathcal{O}_{\text{spec}} : \Omega \longmapsto (\lambda_n(\Omega))_{n \geq 1}$$

is not injective on the class of all planar domains.

Boundary 7.5 (Classical-quantum firewall). *Sharing the domain Ω does not identify the classical collision map with the quantum spectral map. Any semiclassical claim requires an explicit limit, observable class, and theorem. No solution of inverse spectral geometry or quantum chaos is claimed here.*

8 Dissipative and non-specular variants

The reference dynamics is elastic. A normal restitution model with $e \in [0, 1]$, unchanged tangential velocity, and particle mass $m > 0$ gives

$$v_n^+ = -e v_n^-, \quad \Delta E_{\text{diss}} = \frac{m}{2}(1 - e^2)(v_n^-)^2 \geq 0.$$

If $e < 1$, speed is no longer constant, so (s, φ) is not a closed state space: speed or energy must be added. Friction, stochastic reflection, rough walls, moving boundaries, and compliant impact each require a new channel and constitutive law. The invariant measure equation (4) is not inherited automatically.

9 Observation invariants and protocol requirements

Definition 9.1 (Recoverable quantity). Let $\mathcal{O} : X \rightarrow Y$. A quantity $I : X \rightarrow V$ is exactly recoverable from $\mathcal{O}$ if there exists $\tilde{I} : Y \rightarrow V$ such that

$$I = \tilde{I} \circ \mathcal{O}.$$

Equivalently, I is constant on every observation fiber $\mathcal{O}^{-1}(y)$.

Definition 9.2 (Group invariant readout). For a group $G \curvearrowright X$, a readout $I : X \rightarrow V$ is G -invariant when $I(gx) = I(x)$ for all admissible g, x . Equivariance, scale covariance, and estimator stability are not called invariance.

A complete billiard protocol records at least:

1. table class, boundary regularity, corner and grazing policy;
2. boundary parameterization, tangent and normal orientation;
3. speed or energy convention, time unit, and reference frame;

Object	Exact status	Required qualification
Collision measure μ_{∂}	Invariant under regular elastic T	Exclude singular orbit set; declare angle convention
Disk coordinate p	First integral of T_R	Disk geometry and elastic reflection
Finite symbolic word	Deterministic coarse readout	Partition, horizon, boundary rule, missing events
Marginal entropy	Channel statistic	Log base, positive counts, not KS entropy
Laplacian spectrum	Rigid-motion invariant; scale covariant	Boundary condition, multiplicity, operator convention
$ \Omega \lambda_n$	Similarity invariant in two dimensions	Area and eigenvalue belong to the same domain

Table 1: No quantity is called an observation invariant without its channel and transformation class.

4. collision section, roof-time treatment, and observation horizon;
5. deterministic readout, partition, spatial and temporal resolution;
6. missing-impact, duplicate-impact, and misclassification policy;
7. stochastic kernel or explicit noiseless assumption;
8. log base, metric, regression window, weights, and zero guards;
9. spectral boundary condition, multiplicity convention, mass, and $\hbar$ convention whenever energy is reported;
10. estimator, uncertainty model, and declared exactness class.

10 Reproducible benchmarks

The release includes executable tests and a machine-readable benchmark table. The exact reference cases are:

1. normalization of μ_{∂} and the change $p = \sin \varphi$;
2. invertibility, conserved p , roof time, and rotation number for the disk map;
3. reconstruction of p_n from two consecutive impacts in a strictly convex disk;
4. two cyclic binary words with identical marginal entropy but different conditional transition entropy;
5. exact rectangular Dirichlet eigenvalues, a^{-2} spectral scaling, and invariance of $|\Omega|\lambda_n$;
6. the dimensional bridge $E_n = \hbar^2 \lambda_n / (2m)$;
7. finite-word non-injectivity and zero-denominator guards.

For the declared disk benchmark $R = 2.3 \text{ m}$, $p = 0.4$, and $c_0 = 3 \text{ m s}^{-1}$,

$$\rho = \frac{\arccos(0.4)}{\pi} \approx 0.36901012, \quad \tau \approx 1.40532321 \text{ s}.$$

For the cyclic words 01010101 and 00110011, both marginal entropies equal one bit, while their empirical first-order conditional entropies are respectively zero and one bit. This is an exact order-sensitivity benchmark, not a chaos diagnostic.

11 Claim register and boundary

Claim	Status	Scope
Collision-map measure preservation	Standard proposition	Regular elastic billiard, singular sets excluded
Two-impact reconstruction	Proposition	Known strictly convex C^1 table, consecutive regular impacts
Finite symbolic non-injectivity	Proposition	Finite alphabet and horizon on uncountable state set
Disk collision formulas	Theorem	Circular table with declared coordinates
Weyl area recovery	Standard theorem	Planar finite-area Dirichlet Laplacian
Isospectral non-identifiability	Standard theorem	General planar-domain class
Positive finite-time slope implies chaos	Rejected	No such implication
Spectrum determines arbitrary shape	Rejected	False on general planar domains
New billiard or quantum-chaos theorem	Rejected	Not claimed

The admissible program-level conclusion is limited:

Mathematical billiards form a controlled laboratory in which continuous flow, collision reduction, deterministic projection, finite symbolic recording, and spectral observation can be compared without identifying them. Recoverability is a property of the declared channel and hypothesis class, not of the word “observation” alone.

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